








Advanced Time Series Analysis and Forecasting

Chapter 2: Structural breaks and nonlinear models



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Acronyms used in this material (1/3)

- **ADF**: Augmented Dickey–Fuller (test)
- **AI**: Artificial Intelligence
- **AIC**: Akaike Information Criterion
- **AR**: AutoRegressive (model); in event studies: Abnormal Return
- **ARCH**: AutoRegressive Conditional Heteroskedasticity
- **ARDL**: AutoRegressive Distributed Lag (model)
- **BDS**: Brock–Dechert–Scheinkman (test)
- **BIC**: Bayesian Information Criterion
- **BLS**: Bureau of Labor Statistics (US)
- **BNR**: Banca Națională a României (the National Bank of Romania)
- **CI**: Confidence Interval
- **COVID-19**: COronaVIrus Disease 2019
- **CPI**: Consumer Price Index
- **CRPS**: Continuous Ranked Probability Score
- **CSW**: Chu, Stinchcombe and White (monitoring boundary)
- **CUSUM**: CUmulative SUM
- **DM**: Diebold–Mariano (test)
- **ESTAR**: Exponential Smooth Transition AutoRegression
- **FCLT**: Functional Central Limit Theorem
- **FRED**: Federal Reserve Economic Data
- **GARCH**: Generalised AutoRegressive Conditional Heteroskedasticity

Acronyms used in this material (2/3)

- **GDP**: Gross Domestic Product
- **GW**: Giacomini–White (test of conditional predictive ability)
- **HAC**: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- **HICP**: Harmonised Index of Consumer Prices
- **HLN**: Harvey–Leybourne–Newbold (small-sample correction of the DM test)
- **ICSS**: Iterated Cumulative Sums of Squares
- **IFS**: International Financial Statistics (IMF)
- **IMF**: International Monetary Fund
- **JAE**: Journal of Applied Econometrics
- **KSS**: Kapetanios, Shin and Snell (unit-root test against ESTAR)
- **LLM**: Large Language Model
- **LM**: Lagrange Multiplier
- **LM1**: Lagrange Multiplier linearity test, first-order expansion
- **LM3**: Lagrange Multiplier linearity test, third-order expansion
- **LR**: Likelihood Ratio (test)
- **LS**: Least Squares
- **LSTAR**: Logistic Smooth Transition AutoRegression
- **LWZ**: Liu, Wu and Zidek (modified Schwarz criterion)
- **MFM**: Modelling Financial Markets

Acronyms used in this material (3/3)

- **MOSUM**: MOving SUM (of residuals)
- **MPQ**: McConnell and Perez-Quiros (2000)
- **MSFE**: Mean Squared Forecast Error
- **OECD**: Organisation for Economic Co-operation and Development
- **OLS**: Ordinary Least Squares
- **OLS-CUSUM**: CUSUM of OLS residuals
- **ONS**: Office for National Statistics (UK)
- **PELT**: Pruned Exact Linear Time (change-point algorithm)
- **PPP**: Purchasing Power Parity
- **RMSE**: Root Mean Squared Error
- **ROBOR**: Romanian Interbank Offer Rate
- **SCA**: Seasonally and Calendar Adjusted
- **SETAR**: Self-Exciting Threshold AutoRegression
- **SSR**: Sum of Squared Residuals
- **STAR**: Smooth Transition AutoRegression
- **SVAR**: Structural Vector AutoRegression
- **TAR**: Threshold AutoRegression
- **TSA**: Time Series Analysis
- **UK**: United Kingdom
- **US**: United States
- **VAR**: Vector AutoRegression
- **VECM**: Vector Error Correction Model

Today's question and route

- **Question**: when do the parameters of a time-series model change, and when is the dynamics itself state dependent?
  - a **break** is a change in the parameters at some dates; a **nonlinear** model lets the parameters depend on the past of the series
  - both produce the same symptoms: unstable forecasts, spurious persistence, rejected linearity
- **Route** of the chapter
  - breaks: Chow, sup-Wald (Andrews), Bai–Perron, monitoring (CUSUM, CSW), variance breaks (ICSS), unit roots with breaks
  - forecasting under breaks: the choice of the estimation window
  - nonlinear models: TAR/SETAR, STAR (LSTAR, ESTAR), nonlinearity tests, nonlinear forecasts
- We build on TSA, Chapter 3 (Perron (1989), Zivot and Andrews (1992)) and on Chapters 0 and 1 (HAC, bootstrap, DM tests); Seminar 2 comes before this lecture

Learning outcomes

- Derive the Chow test and explain why searching over the break date changes its null distribution (the Davies problem)
- Test for one or several breaks with unknown dates (sup-, exp-, ave-Wald, Bai–Perron), select their number and build confidence intervals for the dates
- Monitor a model in real time with a controlled false-alarm rate, and detect breaks in variance
- Choose the estimation window when the data contain breaks, and evaluate the choice out of sample
- Specify, test, estimate and forecast threshold and smooth-transition models, with valid inference when the threshold is not identified under the null

Reading, data and tools

- Surveys: Casini and Perron (2019), Hansen (2001), Rossi (2013), van Dijk, Teräsvirta and Franses (2002), Hansen (2011)
  - textbooks: Hamilton (1994), Ch. 22 (regime changes); Kilian and Lütkepohl (2017); the forecasting survey Petropoulos et al. (2022) (open access); bridge: Huang and Petukhina (2022)
- Python Quantlets of this chapter: Quantlets/Ch_02
  - Bai–Perron dynamic programming, sup-Wald, ICSS, CSW monitoring, Hansen's bootstrap and LR interval, LSTAR and ESTAR by nonlinear least squares, all written out in `numpy/scipy`
  - R users: the `strucchange` package Zeileis et al. (2002) implements the same break tests
- Lecture notebook: open in Google Colab

Data used in this chapter

Series	Source	Use
US ex-post real interest rate, 1961–1986	JAE data archive (Bai–Perron); rebuilt from FRED (TB3MS, CPIAUCNS)	Bai–Perron
US real GDP growth, quarterly	FRED (GDPC1)	variance break
HICP inflation, Romania (y/y and m/m)	Eurostat (prc_hicp_minr)	breaks, monitoring
EUR/RON, daily	BNR reference rate	ICSS
Real GDP, Romania, quarterly	Eurostat (namq_10_gdp)	unit roots
Unemployment rate, US men aged 20+	BLS via FRED (LNS13000025 / LNS11000025; not adjusted: LNU03000025 / LNU01000025)	TAR, LSTAR
Real dollar-sterling rate	FRED (DEXUSUK, CPIAUCNS), OECD and ONS UK CPI	ESTAR
Canadian lynx trappings, 1821–1934	R data set read by statsmodels	SETAR

- All sources are public and need no account or key; monthly data up to August–September 2026
- HICP: harmonised index of consumer prices; BLS: US Bureau of Labor Statistics; ONS: UK Office for National Statistics; JAE: Journal of Applied Econometrics

Origins of break tests and threshold models



Gregory C. Chow

Portrait: unknown author (1950s–1960s); public domain; Wikimedia Commons



Howell Tong

Photo: Mr and Mrs Howell Tong / IMS; CC BY 3.0; Wikimedia Commons

- 1960: Chow (1960) tests whether two regressions share their coefficients; the same year Quandt (1960) maximises the likelihood ratio over the switch date
- 1980: Tong and Lim (1980) propose threshold autoregression: a linear model in each regime, nonlinear as a whole, able to produce limit cycles

Setting and notation

- Linear model $y_t = x_t'\beta_t + u_t$, $t = 1, \ldots, T$; stability means H_0: $\beta_t = \beta$ for all t
  - x_t: the $k \times 1$ vector of regressors; β_t: the coefficients at date t; u_t: the error
  - one break at $T_1 = [\pi T]$ ($[\cdot]$: the integer part): $\beta_t = \beta_1$ for $t \leq T_1$ and $\beta_t = \beta_2$ after; $\pi \in (0, 1)$ is the **break fraction**
  - m breaks $T_1 < \cdots < T_m$ split the sample into $m + 1$ regimes
- Kinds of change
  - **pure** change (all coefficients) or **partial** change ($y_t = x_t'\beta + z_t'\delta_j + u_t$: only δ moves)
  - z_t: the regressors whose coefficients change; δ_j: their value in regime j; β: the coefficients common to all regimes
  - in the mean, in the dynamics, in the variance; abrupt or gradual (random-walk coefficients, Stock and Watson (1996))
- In TSA, Chapter 3 a break was a nuisance for unit-root tests; here the break is the object of inference

The Chow test (1/2)

- Known date T_1, k regressors, i.i.d. Normal errors Chow (1960)
  - SSR: the sum of squared residuals; S_0: of the pooled regression on all T observations; S_1, S_2: of the regressions before and after T_1
- The test: how much the fit improves when each sub-sample has its own coefficients
  - $F = \dfrac{(S_0 - S_1 - S_2)/k}{(S_1 + S_2)/(T - 2k)} \sim F(k, T - 2k)$ under H_0
  - $S_0 - S_1 - S_2 \geq 0$: the reduction of the SSR, per extra coefficient; a large F is evidence of a break
  - equivalent to the F test of $\delta = 0$ in $y_t = x_t'\beta + (x_t 1\{t > T_1\})'\delta + u_t$; δ: the change of the coefficients after T_1

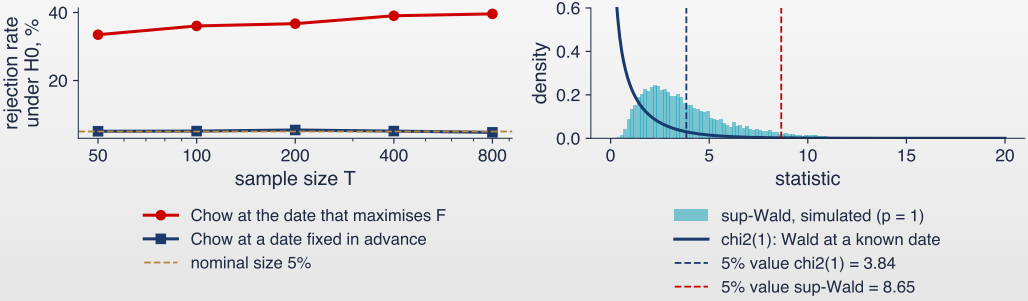
The Chow test (2/2)

- Robust version: the Wald statistic $W_T(T_1) = \hat{\delta}' \hat{V}_{\delta}^{-1} \hat{\delta}$, asymptotically $\chi^2(k)$
  - $\hat{V}_{\delta}$: a heteroskedasticity- or HAC-robust covariance matrix of $\hat{\delta}$ (Chapter 0)
  - the classical F also assumes equal variances in the two regimes
- **Predictive** Chow test when $T - T_1 < k$ (too few observations after the break to estimate the model):
  - $F = \dfrac{(S_0 - S_1)/(T - T_1)}{S_1/(T_1 - k)}$
  - it asks whether the model estimated up to T_1 forecasts the last observations; the ancestor of forecast-breakdown tests (Section 7)

The date chosen by looking at the data

- In practice T_1 is rarely known: the researcher sees a jump in the plot and tests there
  - the test then uses the date that *maximises* the statistic: a data-snooping problem (Chapter 0)
- Under H_0 the break date is **not identified**: the parameter π appears only under the alternative Davies (1987)
  - standard asymptotics fail; the null distribution of $\sup_\pi W_T(\pi)$ is that of the supremum of a stochastic process
- The same problem returns for the threshold of a TAR model (Section 8) and for the transition parameter of a STAR model (Section 9)

A Chow test at the data-chosen date



⊡ Left: white noise, mean-shift Chow test at 5%, 4 000 replications per T; right: simulated null density of $\sup_\pi W_T(\pi)$, $\pi \in [0.15, 0.85]$, against $\chi^2(1)$

ATS_ch2_chow_snooping

Interpreting the size distortion

- At a fixed date the test keeps its size (5.1% for $T = 100$); at the maximising date it rejects a true null in 33% to 40% of samples
  - the distortion grows slowly with T: more candidate dates, a larger maximum
- The 5% critical value of the supremum is 8.65, against 3.84 for $\chi^2(1)$; with two coefficients 11.69
- Lesson: the date search is part of the test; the critical values must account for it

Recap: Instability and known dates

- A break is a change of β_t; pure or partial, in mean, dynamics or variance
- Chow: an F test valid only for a date fixed before seeing the data
- Choosing the date from the data multiplies the size several times
- The break date is a nuisance parameter not identified under H_0

The sup-Wald test of Andrews (1993) (1/2)

- Compute the Chow–Wald statistic $W_T(\pi)$ at every candidate break fraction and take the largest Andrews (1993)
  - $\sup_{\pi \in \Pi} W_T(\pi), \quad \Pi = [\pi_0, 1 - \pi_0]$
  - Π: the candidate fractions; π_0: the **trimming**, usually 0.15, so that each regime has at least 15% of the sample
  - LM, Wald and LR versions share the same limit; Quandt (1960) used the LR version
- **Theorem** Andrews (1993): under H_0 and regularity conditions (stationary, mixing regressors)
  - $W_T(\cdot) \Rightarrow Q_p(\pi) = \dfrac{\|B_p(\pi) - \pi B_p(1)\|^2}{\pi(1-\pi)}$
  - $\Rightarrow$: convergence of the whole process in π; B_p: a p-dimensional standard Brownian motion; p: the number of coefficients allowed to break
  - $B_p(\pi) - \pi B_p(1)$ is a Brownian bridge; $\pi(1-\pi)$ is its variance: $Q_p(\pi)$ is a squared, standardised bridge (▸ Appendix)

The sup-Wald test of Andrews (1993) (2/2)

- Limit: $\sup W_T \to_d \sup_{\pi \in \Pi} Q_p(\pi)$
  - at a fixed π, $Q_p(\pi) \sim \chi^2(p)$; the supremum over many π is much larger
- Why trimming: $\sup_{\pi \in (0,1)} Q_p(\pi) = \infty$ almost surely (law of the iterated logarithm)
  - critical values depend on p and π_0; here simulated: $p = 1$: 7.11 (10%), 8.65 (5%), 12.17 (1%)

Optimal tests: Andrews and Ploberger (1994)

- The supremum is not the only functional; averaging is optimal against alternatives close to H_0 Andrews and Ploberger (1994)
  - $|\Pi| = 1 - 2\pi_0$: the length of the set of candidate fractions; both statistics average over it
  - $\mathrm{expW} = \ln\left(\frac{1}{|\Pi|}\int_{\Pi} \exp\left(\frac{1}{2}W_T(\pi)\right)d\pi\right)$: weighted average power, for medium and large breaks
  - $\mathrm{aveW} = \frac{1}{|\Pi|}\int_{\Pi} W_T(\pi)\,d\pi$: optimal for very small breaks, close to the Nyblom test of random-walk coefficients
- 5% values for $p = 1$, $\pi_0 = 0.15$ (simulated): exp 2.02, ave 2.87; sup: previous slide
- The sup-Wald test has power against *any* kind of instability, not only a single break
  - a rejection says "unstable", not "one break at $\hat{\pi}$"; the number of breaks is the subject of Section 3

Estimating the break date (1/2)

- Least squares: choose the date that minimises the total SSR of the two regimes
  - $\hat{T}_1 = \arg\min_{T_1} [S_1(T_1) + S_2(T_1)]$
  - $S_1(T_1)$, $S_2(T_1)$: the SSR before and after a candidate date T_1; equivalently the date of the largest Wald statistic under homoskedasticity
- Precision Bai (1997)
  - for a fixed break size, $\hat{T}_1 - T_1 = O_p(1)$: the **date** error is bounded in probability and does not grow with T
  - so the fraction $\hat{\pi} = \hat{T}_1 / T$ converges at rate T, faster than the usual $\sqrt{T}$
  - the regime coefficients are $\sqrt{T}$-consistent and asymptotically Normal as if the date were known

Estimating the break date (2/2)

- Confidence interval for the date Bai (1997): with a shrinking break $\Delta_T \to 0$
  - $\Delta_T^2 \sigma^{-2}(\hat{T}_1 - T_1) \to_d \arg\max_s Z(s), \quad Z(s) = W(s) - |s|/2$
  - Δ_T: the size of the change in the mean; σ^2: the error variance; $W(s)$: a two-sided Brownian motion, $s \in \mathbb{R}$
  - the interval is the date $\pm$ quantiles of $\arg\max Z$, rescaled by $\sigma^2/\hat{\Delta}^2$
- Reading
  - large breaks, small noise: a short interval; the interval is asymmetric when variances differ across regimes

Case study: McConnell and Perez-Quiros (2000)

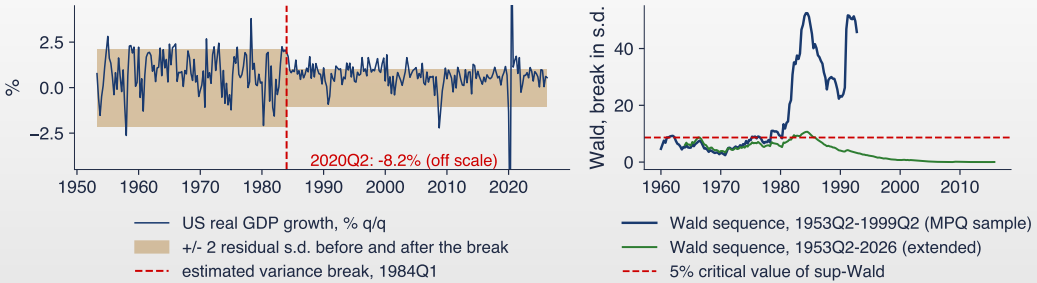


Gasoline queue, Maryland, 15 June 1979

Photo: Warren K. Leffler (1979), Library of Congress; public domain; Wikimedia Commons

- The paper: US real GDP growth, 1953Q2–1999Q2; AR(1) for the conditional mean; then a break in the mean of $\sqrt{\pi/2}\,|\hat{e}_t|$, an unbiased estimate of the standard deviation under Normality McConnell and Perez-Quiros (2000)
  - $\hat{e}_t$: the AR(1) residual; here $\pi = 3.14\ldots$; if $e_t \sim N(0, \sigma^2)$, $\mathbb{E}|e_t| = \sigma\sqrt{2/\pi}$, hence the factor
  - tests: sup-, exp-, ave-Wald Andrews (1993), Andrews and Ploberger (1994); their result: a variance break in 1984Q1, the "Great Moderation"
- Our replication: the same two steps on FRED GDPC1 (today's vintage), robust Wald statistics, simulated p-values; then the sample extended to 2026
- Question: does the variance break survive revised data and the COVID-19 quarters?

The Great Moderation in US output growth



- Top: growth with ± 2 residual standard deviations before and after the estimated break; bottom: the Wald sequence for a break in $\mathbb{E}\sqrt{\pi/2}|e_t|$ (HAC), two samples

ATS_ch2_great_moderation

Interpreting the variance break

- MPQ sample ($T = 185$): sup-Wald 52.5 ($p < 0.001$), exp 23.2, ave 17.5; date 1984Q1, the date found by McConnell and Perez-Quiros (2000)
  - residual s.d. 1.06 before, 0.45 after: the variance falls by a factor of 5.5; no break in the mean equation (sup 3.2, p 0.861)
- To 2019: the break moves by one quarter (1984Q2), variance ratio 4.1; the Great Moderation survived the 2008–2009 recession
- To 2026: sup-Wald falls to 10.7 (p 0.019) and the ratio to 1.1: two outliers (2020Q2: -8.2%) dominate the post-break variance
  - a second break, or outliers? A one-break test cannot say; robust scale estimators or a second break are needed

Recap: One break at an unknown date

- sup-Wald: the supremum of a squared Brownian bridge; trim, and use its own critical values
- exp- and ave-Wald are optimal for moderate and small breaks
- The date is estimated precisely, the CI comes from $\arg\max_s \{W(s) - |s|/2\}$
- Variance breaks: test the mean of $|\hat{e}_t|$; outliers can mask or mimic them

The multiple-break model

- $y_t = x_t'\beta + z_t'\delta_j + u_t$, $t = T_{j-1}+1, \ldots, T_j$, $j = 1, \ldots, m+1$, with $T_0 = 0$, $T_{m+1} = T$ Bai and Perron (1998)
  - $T_1 < \cdots < T_m$: the break dates; δ_j: the coefficients of z_t in regime j; β: those of x_t, common to all regimes
  - pure change: no x_t; mean-shift model: $z_t = 1$
  - minimum segment length $h = [\varepsilon T]$, usually $\varepsilon = 0.15$; errors may be autocorrelated and heteroskedastic across regimes
- Estimator: the **global** minimiser of the SSR over all admissible partitions $(T_1, \ldots, T_m)$
  - break fractions consistent at rate T, as for one break; the dates are estimated jointly, not one at a time
- Brute force needs $O(T^m)$ regressions; dynamic programming needs $O(T^2)$ Bai and Perron (2003)

Dynamic programming

- Step 1: compute $\mathrm{SSR}(i, j)$ of one regression on every segment $[i, j]$ with $j - i + 1 \geq h$ (recursive residuals: $O(T^2)$ operations)
- Step 2: Bellman recursion for the best r-break partition of the first n observations
  - $\mathrm{SSR}(\{T_{r,n}\}) = \min_{rh \leq j \leq n-h} \left[\mathrm{SSR}(\{T_{r-1,j}\}) + \mathrm{SSR}(j+1, n)\right]$
  - $\{T_{r,n}\}$: the best partition of observations $1, \ldots, n$ with r breaks; j: the date of its last break; h: the minimum segment length
  - the optimal m-break partition contains optimal partitions of its prefixes (Bellman's principle)
- One pass gives the optimal partitions for every $m \leq M$: the input of all tests and criteria
  - partial change (β common) needs an iteration between β and the partition Bai and Perron (2003); Seminar 2, A3 runs the recursion by hand

Testing and selecting the number of breaks (1/2)

- $\sup F_T(k)$: no break against exactly k breaks, at the SSR-optimal partition, with robust variance Bai and Perron (1998)
  - M: the largest number of breaks considered (here 5)
  - **UDmax** $= \max_{k \le M} \sup F_T(k)$: no break against an unknown number of breaks, up to M
  - **WDmax**: the same, with each k weighted by the ratio of critical values
- **Sequential** $\sup F_T(\ell + 1 \mid \ell)$: ℓ breaks against $\ell + 1$
  - for each of the $\ell + 1$ segments, the largest single-break statistic; add a break while it rejects
  - recommended strategy Bai and Perron (2003): UDmax or WDmax first, then the sequential tests

Testing and selecting the number of breaks (2/2)

- Information criteria: pick the m with the smallest value
  - $\mathrm{BIC}(m) = \ln(\mathrm{SSR}_m/T) + p^* \ln T/T$
  - **LWZ**: $\ln \frac{\mathrm{SSR}_m}{T-p^*} + p^* \frac{0.299}{T} (\ln T)^{2.1}$ Liu, Wu and Zidek (1997)
  - SSR_m: the SSR of the best m-break partition; p^*: the number of coefficients and break dates; the second term penalises extra breaks
- Known weaknesses
  - BIC picks too many breaks under serial correlation, LWZ too few with small breaks

Case study: Bai and Perron (2003), the US real interest rate

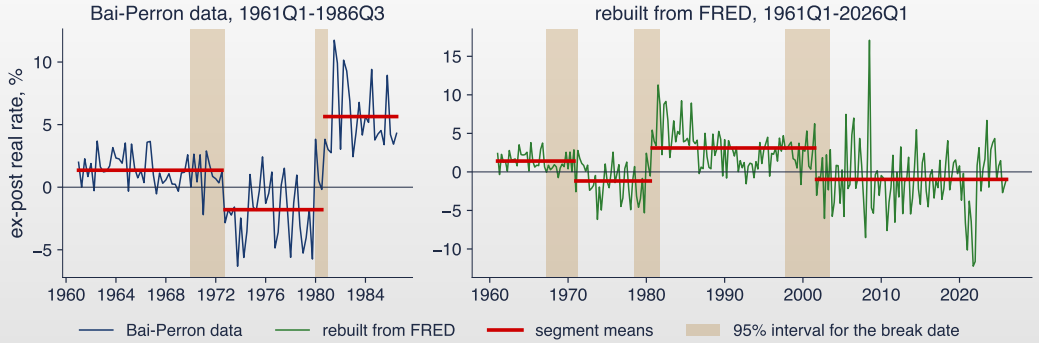


Paul Volcker (left) and Ronald Reagan, December 1981

Photo: White House Photographic Office (1981); public domain; Wikimedia Commons

- The paper, Section 6.1: the ex-post real rate of Garcia and Perron (1996), quarterly 1961Q1–1986Q3 ($T = 103$), mean-shift model, $\varepsilon = 0.15$ ($h = 15$), $M = 5$, HAC variances that differ across segments Bai and Perron (2003)
  - our replication uses their data file (JAE data archive) and the same settings
- Extension: the series rebuilt from FRED (T-bill rate at the end of the quarter minus next-quarter CPI inflation), correlation 0.99 with the original, 1961Q1–2026Q1
- Question: how many regimes, and do they match the history of monetary policy?

Mean shifts in the US real interest rate



- Red: segment means of the BIC/LWZ partition; shaded: 95% intervals for the dates Bai (1997), heterogeneous long-run variances. Top: original data; bottom: rebuilt to 2026

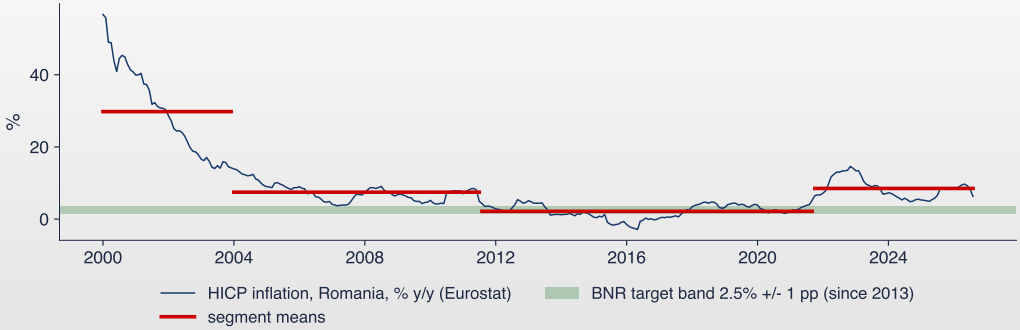
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Interpreting the Bai–Perron results

m	0	1	2	3	4	5
SSR	1215	645	456	445	–	–
$\sup F_T(m)$	–	61.1	46.1	36.1	27.4	20.6
5% value	–	8.74	7.27	6.05	–	–
BIC	2.513	1.970	1.713	1.779	–	–
LWZ	2.550	2.082	1.901	2.043	–	–

- UDmax 61.1 (5%: 8.87); sequential: $F(2|1) = 36.9$, $F(3|2) = 16.1$, $F(4|3) = 0.04$ (5%: 10.54, 11.56): 3 breaks, 1966Q4, 1972Q3, 1980Q3
  - BIC and LWZ choose 2 breaks: 1972Q3, 1980Q3; the dates are the ones reported by Bai and Perron (2003)
- Means 1.36%, -1.80%, 5.64%: negative real rates in the 1970s, then the 1979–1981 disinflation; 95% intervals [1970Q1, 1972Q4], [1980Q1, 1981Q1]
- Extended sample: 3 breaks (1970Q4, 1980Q3, 2001Q3); the last regime has a mean of -0.97%: the low-rate era after 2001, interval [1997Q4, 2003Q3]

Mean shifts in Romanian inflation



- HICP inflation, y/y, January 2000–August 2026 ($T = 320$), Bai–Perron mean shifts, $\varepsilon = 0.15$, $M = 5$; red: the partition chosen by BIC and LWZ

ATS_ch2_ro_inflation_breaks

Interpreting the Romanian inflation regimes

- BIC and LWZ: 3 breaks (December 2003, July 2011, September 2021); means 29.8%, 7.5%, 2.2%, 8.5%
  - disinflation, inflation targeting (adopted in August 2005), the low-inflation decade, the 2021–2023 surge
- The robust sequential procedure stops at 1 break: $\sup F(1) = 12.8$, but $F(2|1) = 2.06$
  - inflation is very persistent within regimes: the HAC variance absorbs the shifts, and the test loses power
  - for the same reason the 95% intervals for the dates span years (second date: [January 2000, December 2014]); they are not drawn
- Better model: partial change in the intercept of an AR model, so that the errors are close to white noise (project idea)

Change-point detection in statistics and machine learning

- Penalised segmentation: minimise $\sum_j \mathrm{cost}(\text{segment } j) + \beta\, m$ over partitions
  - cost: e.g. the SSR or minus the log-likelihood of a segment; $\beta > 0$: the penalty per break; m: the number of breaks
  - PELT prunes the Bellman recursion to $O(T)$ under mild conditions Killick, Fearnhead and Eckley (2012); binary segmentation and wild binary segmentation Fryzlewicz (2014) scale to long series
  - the Python package `ruptures` implements them Truong, Oudre and Vayatis (2020); the BIC of Bai–Perron is a particular penalty
- What econometrics adds: inference on the number of breaks and confidence intervals for dates under serial correlation
  - common breaks in systems of equations Oka and Perron (2018); survey Casini and Perron (2019)
- Engineering practice tunes the penalty on labelled data; econometrics fixes the size of a test: both are valid if the choice is pre-registered

Recap: Multiple breaks

- Global least squares over all partitions, by dynamic programming in $O(T^2)$
- UDmax/WDmax, then sequential $F(\ell+1|\ell)$; BIC and LWZ as a cross-check
- US real rate: breaks in 1972Q3 and 1980Q3 replicated; a further regime after 2001
- Persistent errors: robust tests lose power and date intervals become wide

CUSUM and MOSUM tests (1/2)

- Recursive residuals Brown, Durbin and Evans (1975): the standardised one-step forecast error of the model estimated up to $t-1$
  - $w_t = (y_t - x_t'\hat{\beta}_{t-1})/\sqrt{1 + x_t'(X_{t-1}'X_{t-1})^{-1}x_t}$
  - $\hat{\beta}_{t-1}$: OLS on observations $1, \ldots, t-1$; X_{t-1}: the matrix of their regressors; the root rescales for estimation uncertainty
  - under H_0 with Normal errors, w_t are i.i.d. $N(0, \sigma^2)$
- CUSUM: the cumulated standardised recursive residuals
  - $W_r = \hat{\sigma}^{-1} \sum_{t=k+1}^{r} w_t$, 5% boundary $\pm 0.948[\sqrt{T-k} + 2(r-k)/\sqrt{T-k}]$
  - k: the number of regressors (the first k observations start the recursion); W_r behaves like a Brownian motion; a crossing signals a break

CUSUM and MOSUM tests (2/2)

- What CUSUM detects
  - changes in the intercept and a systematic drift of forecast errors; weak against changes that average out
- OLS-CUSUM uses ordinary residuals: a Brownian bridge limit Ploberger and Krämer (1992); MOSUM sums over a moving window of fixed width Chu, Hornik and Kuan (1995)
  - MOSUM reacts faster to a break in the middle of the sample and to temporary changes
- These are **retrospective** tests: the whole sample is available when the test is run

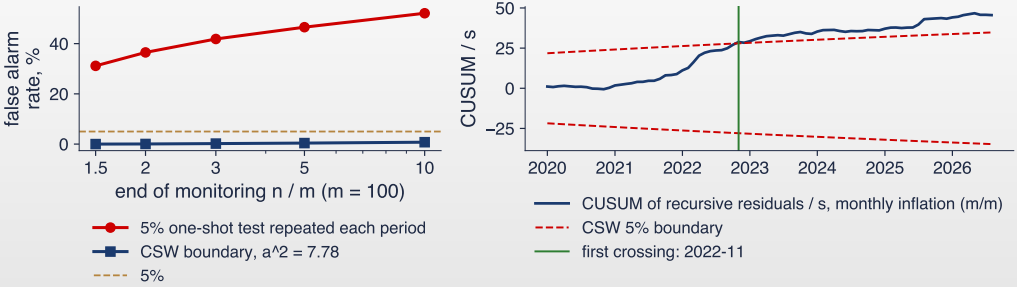
Monitoring: the failure of repeated tests (1/2)

- A central bank estimates a model on m historical observations and asks, at every new release, "has it broken down?"
  - a 5% test repeated at $n = m+1, m+2, \ldots$ rejects a true H_0 eventually with probability 1: the law of the iterated logarithm
- Chu, Stinchcombe and White (1996): monitor the CUSUM of the new recursive residuals and stop the first time it crosses a widening boundary
  - $Q_n = \hat{\sigma}^{-1} \sum_{t=m+1}^{n} w_t$, alarm if $|Q_n| > \sqrt{n\,[a^2 + \ln(n/m)]}$
  - n: the current sample size; a: a constant chosen to fix the false-alarm probability; the $\ln(n/m)$ term widens the boundary as time passes

Monitoring: the failure of repeated tests (2/2)

- The boundary comes from Robbins and Siegmund (1970):
  - $P\{\exists u \geq 1 : |W(u) - W(1)| \geq \sqrt{u(a^2 + \ln u)}\} = 2[1 - \Phi(a) + a\varphi(a)]$
  - W: a standard Brownian motion; $u = n/m$; Φ, φ: the standard Normal distribution and density functions
  - $a^2 = 7.78$: size 5.1% over an *infinite* horizon; $a^2 = 6.25$: 10.0%
- Forecast-based monitoring: forecast breakdowns Giacomini and Rossi (2009) and the fluctuation test of relative accuracy Giacomini and Rossi (2010) (Chapter 1)

False alarms and real-time monitoring of Romanian inflation



⊡ Left: white noise, $m = 100$, 2 000 replications; right: mean of monthly HICP inflation (m/m), historical sample 2015–2019, monitoring from January 2020

ATS_ch2_monitoring

Interpreting the monitoring results

- Repeated 5% tests raise a false alarm in 36% of samples by $n = 2m$ and in 52% by $n = 10m$; the CSW boundary in 0.8%
  - the CSW size is reached only as $n \to \infty$: over a finite horizon the boundary is conservative
- Romania: historical mean 0.15% per month, $\hat{\sigma} = 0.51$; the CUSUM crosses the boundary in November 2022
  - the repeated naive test would have signalled in January 2022; with a long-run $\hat{\sigma}$ (HAC) the signal comes only in August 2025
- The price of a controlled false-alarm rate is delay: nearly two years after inflation started to rise in 2021

Recap: Monitoring

- CUSUM and MOSUM of recursive residuals: retrospective fluctuation tests
- Repeating a one-shot test as data arrive guarantees a false alarm
- CSW boundary $\sqrt{n[a^2 + \ln(n/m)]}$ controls the size over an infinite horizon
- Romanian inflation: detection in late 2022, a long but honest delay

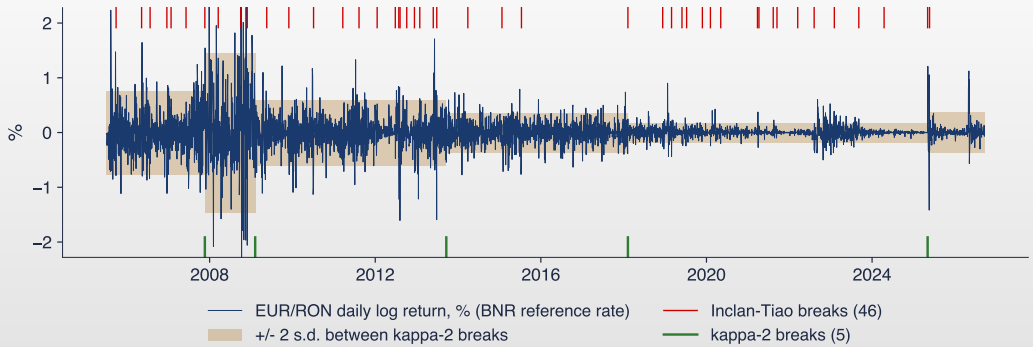
The ICSS algorithm and its correction (1/2)

- Inclán and Tiao (1994): compare the cumulated sum of squares with a straight line
  - $C_k = \sum_{t \le k} a_t^2, \quad D_k = C_k / C_T - k/T, \quad \mathrm{IT} = \sqrt{T/2}\max_k |D_k| \Rightarrow \sup |B^0(r)|$
  - a_t: the demeaned returns; C_k/C_T: the share of the total sum of squares reached at k; with a constant variance it grows like k/T
  - B^0: a Brownian bridge on $[0,1]$; 5% critical value 1.358; the date is the k of the largest $|D_k|$
- ICSS (iterated cumulative sums of squares)
  - apply the test iteratively to sub-segments (binary segmentation), then re-check each break between its neighbours

The ICSS algorithm and its correction (2/2)

- The factor $\sqrt{T/2}$ assumes $\mathrm{Var}(a_t^2) = 2\sigma^4$: i.i.d. Normal data
  - with kurtosis $\kappa = \mathbb{E}a_t^4/\sigma^4$ the statistic is inflated by $\sqrt{(\kappa-1)/2}$; with GARCH, a_t^2 is also autocorrelated
- κ_2 of Sansó, Aragó and Carrion (2004): the same CUSUM, scaled by a long-run variance
  - $\kappa_2 = \max_k |C_k - (k/T)C_T|/\sqrt{T\hat{\omega}_4}$
  - $\hat{\omega}_4$: the long-run variance of $a_t^2 - \hat{\sigma}^2$ (Bartlett kernel, Chapter 0)
  - same limit, valid under fat tails and conditional heteroskedasticity

Variance regimes of EUR/RON



- Daily log returns of the BNR reference rate, 5 352 days; red ticks: ICSS with the Inclán–Tiao statistic; green ticks and shaded bands: ICSS with κ_2

ATS_ch2_variance_breaks

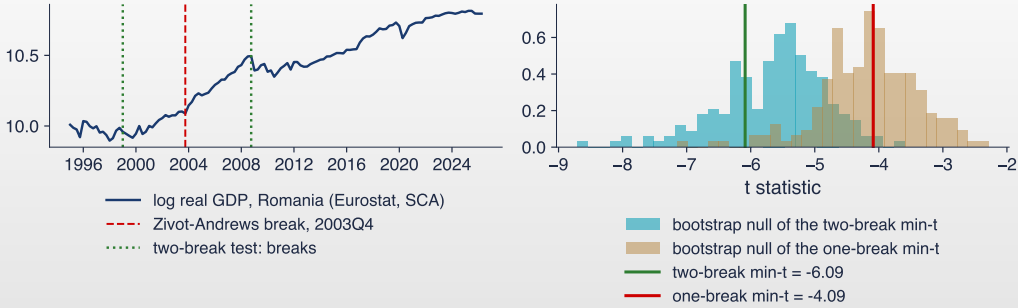
Interpreting the EUR/RON variance breaks

- Excess kurtosis 15.1: the Inclán–Tiao statistic is 24.0 and ICSS finds 46 breaks; $\kappa_2 = 4.69$ and 5 breaks
  - most Inclán–Tiao "breaks" are volatility clusters, not changes of the unconditional variance
- κ_2 dates: 20 November 2007, 6 February 2009, 19 September 2013, 7 February 2018, 5 May 2025
  - the global financial crisis, the end of its turbulence, the calmer managed float after 2018; daily s.d. between 0.09% and 0.73%
- For risk models (MFM, Chapter 5): a GARCH fitted across a variance break overstates persistence ($\alpha + \beta \to 1$) Diebold and Inoue (2001)

Beyond Zivot–Andrews

- TSA, Chapter 3: a level shift mimics a unit root Perron (1989); Zivot and Andrews (1992) choose the date by minimising the ADF t statistic
  - their null is a unit root *without* a break: a break under the null makes the test reject too often
- Two breaks: Lumsdaine and Papell (1997) minimise the t statistic over two dates; Lee and Strazicich (2003) propose an LM test whose null allows breaks
  - Kim and Perron (2009): pre-test the break, then use a test valid under both hypotheses
- Critical values depend on the model, the trimming and the number of breaks; a sieve bootstrap under the null avoids the tables

Romanian GDP: unit root or broken trend?



- Log real GDP (SCA), 1995Q1–2026Q2; one and two breaks in level and trend; null distributions from a sieve bootstrap (AR on Δy_t, 199 replications)

ATS_ch2_unit_root_breaks

Interpreting the unit-root tests with breaks

- ADF with trend: -2.79 (p 0.201); Zivot–Andrews: -4.09 at 2003Q4 (5%: -5.07; bootstrap p 0.523)
- Two breaks (1999Q1 and 2008Q4): min-$t = -6.09$, bootstrap 5% value -7.16, p 0.266
  - the more breaks we allow, the more negative the critical value: the apparent gain disappears
- With about 30 years of quarterly data a unit root cannot be rejected; the convergence story (fast growth, then a 2009 break) remains a hypothesis

Recap: Breaks in variance and unit roots

- ICSS needs the κ_2 correction for fat tails and GARCH
- EUR/RON: a handful of genuine variance regimes, not dozens
- Unit-root tests with breaks: allow breaks under the null or pre-test them; bootstrap the critical values

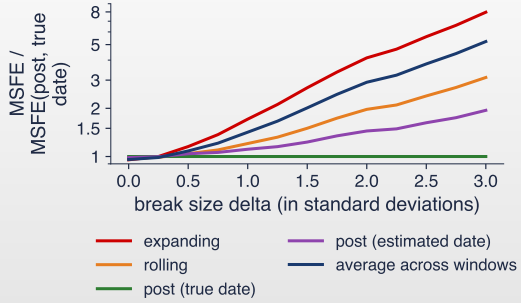
Location shifts cause forecast failure

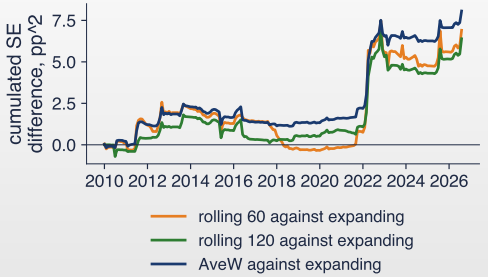
- Clements and Hendry (1998), Clements and Hendry (2006): in equilibrium-correction models, a shift of the equilibrium mean is the main source of systematic forecast errors
  - after the shift, the model keeps pulling the forecast towards the old mean
- Robust devices
  - **intercept correction**: add the last observed error to the forecast
  - **differencing**: forecast Δy; adapts after one period, at the cost of a larger variance
  - **shorter windows**: rolling estimation, or the post-break sample
- Survey of the evidence: Rossi (2013); window choice with time-varying parameters Inoue, Jin and Rossi (2017)

The choice of the estimation window

- Mean shift δ after n_1 pre-break observations, n_2 post-break; forecast with the last $w \geq n_2$ observations Pesaran and Timmermann (2007)
  - $\mathrm{MSFE}(w) = \sigma^2\left(1 + \frac{1}{w}\right) + \left(\frac{(w - n_2)\delta}{w}\right)^2$: variance falls with w, squared bias grows
  - MSFE: the mean squared forecast error of the sample mean over the window; σ^2: the noise variance; $(w - n_2)/w$: the share of pre-break data in the window
  - the optimal window includes *some* pre-break data when δ/σ is small or n_2 short (Seminar 2, A8)
- Feasible rules: estimate the date (Bai–Perron) and use the post-break sample; cross-validate w Pesaran and Timmermann (2007); average forecasts across windows (AveW) Pesaran and Pick (2011)
  - the date is estimated with error exactly when the break is small: post-break windows are then too short

Window choice: simulation and Romanian inflation





- Left: $T = 200$, break $n_2 = 20$ periods before the end, 4 000 replications, MSFE relative to the post-break mean with the true date; right: AR(2), one month ahead, cumulated squared-error gains over the expanding window

ATS_ch2_forecast_windows

Interpreting the window choice

- Simulation: without a break the expanding window wins (0.95); with $\delta = 1$: expanding 1.71, rolling 1.21, AveW 1.42, estimated post-break 1.11
  - $\delta = 2$: 4.13, 1.97, 2.91, 1.44: estimating the date pays when the break is large
- Romania, $P = 200$ months: RMSE expanding 0.664, rolling 60: 0.638, rolling 120: 0.640, AveW 0.633 pp
  - HLN against expanding: AveW 2.20 (p 0.029), rolling 60 1.19 (p 0.234); all gains arrive in 2022, the RMSE in 2021–2023 falls from 0.790 to 0.700
- Averaging across windows is a cheap insurance: never far from the best window, without estimating any date

Recap: Forecasting under breaks

- Location shifts drive forecast failure; intercept correction and differencing make forecasts robust
- Optimal window: a bias–variance trade-off that may include pre-break data
- AveW is robust when the break is uncertain; post-break windows when it is large and well dated
- Evaluate the window choice out of sample with DM–HLN Diebold and Mariano (1995), Harvey, Leybourne and Newbold (1997) (Chapter 1)

Motivation for nonlinear models



Canada lynx

Photo: Michael Zahra (2010); CC BY-SA 3.0; Wikimedia Commons

- Linear models are symmetric: a shock of -1 has the mirror effect of a shock of $+1$, at every state
  - business cycles: unemployment rises fast in recessions and falls slowly in expansions Neftçi (1984)
  - arbitrage: real exchange rates move freely inside a band of transaction costs, revert outside it Michael, Nobay and Peel (1997)
- Ecology: the 9–10 year cycle of the Canadian lynx is asymmetric (slow rise, fast fall); a linear AR cannot produce a stable cycle without noise
- A nonlinear model can: limit cycles, asymmetry, state-dependent persistence and impulse responses

Threshold and self-exciting threshold autoregression (1/2)

- Two-regime **TAR**: an AR model whose coefficients depend on whether q_{t-1} is below or above the threshold γ
  - $y_t = (\phi_{1,0} + \phi_1' \mathbf{y}_{t-1}) 1\{q_{t-1} \leq \gamma\} + (\phi_{2,0} + \phi_2' \mathbf{y}_{t-1}) 1\{q_{t-1} > \gamma\} + \varepsilon_t$
  - $\mathbf{y}_{t-1} = (y_{t-1}, \ldots, y_{t-p})'$: the last p values; $\phi_{j,0}$, ϕ_j: the intercept and AR coefficients of regime j
  - q_{t-1}: the observed threshold variable; γ: the threshold; $1\{\cdot\}$ selects the regime; variances may differ across regimes
- **SETAR** (self-exciting): $q_{t-1} = y_{t-d}$, the series itself with the **delay** lag d
  - notation SETAR($k; p_1, \ldots, p_k$): k regimes with AR orders $p_1, \ldots, p_k$ Tong and Lim (1980)

Threshold and self-exciting threshold autoregression (2/2)

- Dynamics: the **skeleton** $y_t = F(y_{t-1})$, the model with the noise switched off
  - F: the regime-dependent conditional mean; the skeleton can converge to a point, a limit cycle or a chaotic attractor
- Stationarity is a property of the whole model, not of each regime
  - SETAR(2; 1, 1) is ergodic iff $\phi_1 < 1$, $\phi_2 < 1$ and $\phi_1\phi_2 < 1$: a regime may be explosive on its own (Seminar 2, A5)
- Survey of threshold models in economics: Hansen (2011)

Estimation and inference (1/2)

- Concentrated least squares: for each candidate γ (and d) run OLS in both regimes
  - $\hat{\gamma} = \arg\min_{\gamma} S_T(\gamma)$
  - $S_T(\gamma)$: the total SSR given γ; the search runs over the central 70% of the observed values of q
  - $\hat{\gamma}$ is super-consistent, $T(\hat{\gamma} - \gamma) = O_p(1)$, with a nonstandard limit Chan (1993); the slopes are asymptotically Normal as if γ were known
- Testing linearity, H_0: $\phi_1 = \phi_2$: γ is not identified under H_0 (the Davies problem again) Hansen (1996)
  - $\sup_{\gamma} W_T(\gamma)$, with $W_T(\gamma)$ the Wald statistic of $\phi_1 = \phi_2$ at threshold γ, has a limit that depends on the data
  - **fixed-regressor bootstrap**: $y_t^* = \hat{e}_t \eta_t$, $\eta_t \sim N(0,1)$, same regressors, recompute sup W^*; $\hat{e}_t$: the residuals

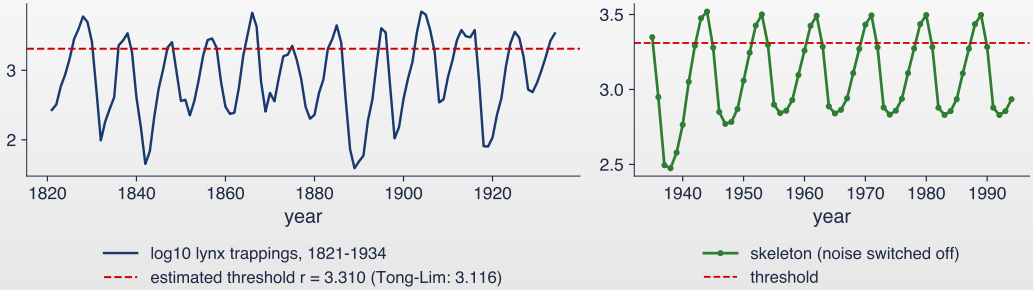
Estimation and inference (2/2)

- Confidence set for the threshold Hansen (2000): all γ with a small likelihood ratio
  - $\mathrm{LR}_T(\gamma) = T\,\dfrac{S_T(\gamma) - S_T(\hat{\gamma})}{S_T(\hat{\gamma})}$
  - the relative increase of the SSR when the threshold is moved from $\hat{\gamma}$ to γ
- Limit distribution and critical value
  - $\xi = \max_s[2W(s) - |s|]$, W a two-sided Brownian motion; $P(\xi \le x) = (1 - e^{-x/2})^2$
  - 95% value $-2\ln(1 - \sqrt{0.95}) = 7.35$: the 95% set is $\{\gamma : \mathrm{LR}_T(\gamma) \le 7.35\}$
  - under heteroskedasticity divide LR by $\hat{\eta}^2$, a variance-ratio correction estimated from the residuals (Appendix)

Case study: Tong and Lim (1980), the Canadian lynx

- The paper, Section 9: $\log_{10}$ of the yearly lynx trappings, 1821–1934; SETAR(2; 7, 2) with $d = 2$, threshold 3.116 Tong and Lim (1980)
  - $d = 2$ has an ecological reading: a lynx is fully grown in the autumn of its second year
- Our replication with the threshold of the paper: lower regime $y_t = 0.546 + 1.032y_{t-1} - 0.173y_{t-2} + \dots$ (seven lags), upper regime $2.345 + 1.533y_{t-1} - 1.276y_{t-2}$
  - the lower regime reproduces the published coefficients to three decimals; the free least-squares threshold is 3.310 (SSR 3.764 against 3.943)
- Question: does the fitted model generate the lynx cycle by itself?

The lynx cycle and the SETAR skeleton



- Left: data and the estimated threshold; right: the skeleton iterated 60 years from the last observations, noise switched off

ATS_ch2_lynx

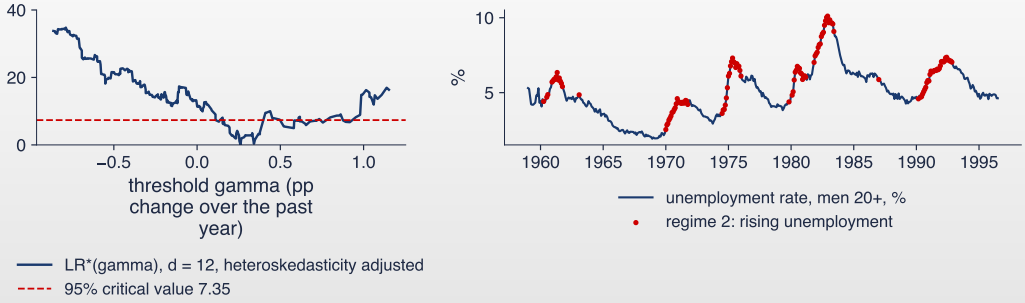
Interpreting the lynx SETAR

- The skeleton settles on a stable limit cycle of period 9 years: the model produces the cycle without any noise
  - a linear AR(11) needs noise to sustain the oscillation; its skeleton converges to the mean
- The rise is slow (lower regime, seven lags), the fall is fast (upper regime, an explosive AR(2) on its own): the asymmetry of the data
- Residual variance 0.0352 against 0.0364 for AR(11): a small in-sample gain, a large gain in the description of the dynamics

Case study: Hansen (1997), US unemployment

- The paper, Section 5: unemployment rate of men aged 20 and over (unemployed / labour force), monthly 1959.1–1996.7; Δy_t on a constant and 12 lags Hansen (1997)
  - y_t: the unemployment rate in %; threshold variable $q_{t-1} = y_{t-1} - y_{t-d}$, $d = 2, \ldots, 12$: the change over the last $d-1$ months; trimming 15%
  - robust sup-Wald test, bootstrap p-values (1000 replications); the LS estimate $\hat{d} = 12$, $\hat{\gamma} = 0.302$, 95% interval [0.213; 0.340], 314 and 124 observations
- Our replication: the same construction from the BLS series on FRED (today's seasonal adjustment), the same model, $d = 12$
- Question: is there a "recession regime" with different dynamics, and how precisely is its threshold estimated?

A threshold model for US unemployment



- Left: heteroskedasticity-adjusted $\mathrm{LR}^*(\gamma)$, $d = 12$; the 95% set is where the curve lies below 7.35. Right: months in the rising-unemployment regime ($q_{t-1} > \hat{\gamma}$)

ATS_ch2_unemp_tar

Interpreting the unemployment TAR

- $\hat{\gamma} = 0.261$ pp (paper: 0.302), regimes of 314 and 124 months, the same split as in Hansen (1997); 95% set [0.131, 0.917]
  - sup-Wald 47.2, bootstrap p 0.015; linearity is rejected at 5% for 10 of the 11 delays
- Rising regime: intercept 0.065, AR(1) 0.18, AR(2) 0.28; other months: -0.019, -0.15, 0.06
  - increases feed on themselves; in expansions unemployment is close to a random walk with a slight downward drift
- With today's data the SSR is smallest at $d = 3$ ($\hat{\gamma} = -0.24$): the delay is fragile, the threshold effect is not; 1959–2019: $\hat{\gamma} = 0.261$, $p < 0.001$

Recap: Threshold autoregression

- A piecewise-linear AR switching on an observed variable; can generate limit cycles
- $\hat{\gamma}$ super-consistent; slopes as if γ were known
- Linearity tests need the bootstrap; threshold intervals invert LR with the value 7.35
- Lynx and unemployment: both landmark results replicate

STAR models (1/2)

- A weighted average of two AR regimes, with a weight G that moves smoothly between 0 and 1 Teräsvirta (1994)
  - $y_t = \phi_1' x_t[1 - G(s_t; \gamma, c)] + \phi_2' x_t G(s_t; \gamma, c) + \varepsilon_t, \quad x_t = (1, y_{t-1}, \ldots, y_{t-p})'$
  - s_t: the transition variable (e.g. a lag of y_t); c: the location of the transition; $\gamma > 0$: its speed
  - $G = 0$: regime 1 with coefficients ϕ_1; $G = 1$: regime 2 with ϕ_2; in between, a mixture
- A continuum of regimes
  - smooth aggregation of many agents with different thresholds, or time aggregation, produces smooth switching

STAR models (2/2)

- **LSTAR** (logistic): low against high values of s_t
  - $G = [1 + \exp\{-\gamma(s_t - c)/\hat{\sigma}_s\}]^{-1}$
  - $\gamma \to \infty$ gives the TAR, $\gamma \to 0$ the linear AR; $\hat{\sigma}_s$: the standard deviation of s_t, which makes γ scale free and easier to estimate
- **ESTAR** (exponential): small against large *deviations* from c, symmetric
  - $G = 1 - \exp\{-\gamma(s_t - c)^2\}$
- Survey with the modelling cycle (specification, estimation, evaluation): van Dijk, Teräsvirta and Franses (2002)

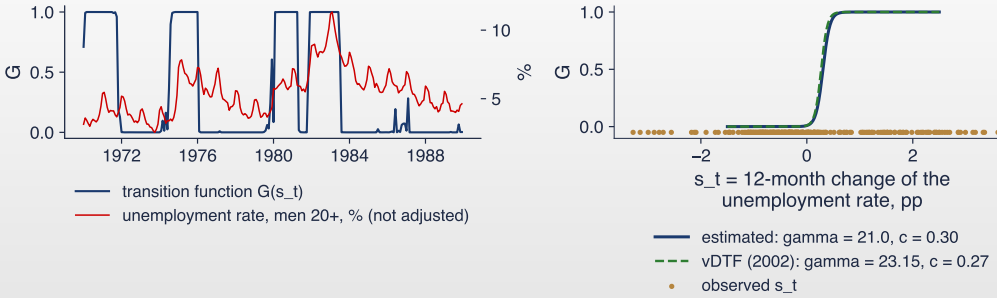
Testing linearity against STAR

- Under H_0: $\gamma = 0$, the parameters c and ϕ_2 are not identified: replace G by a third-order Taylor expansion around $\gamma = 0$ Luukkonen, Saikkonen and Teräsvirta (1988)
  - auxiliary regression $\hat{\varepsilon}_t = \beta_0' \mathrm{x}_t + \beta_1' \tilde{\mathrm{x}}_t s_t + \beta_2' \tilde{\mathrm{x}}_t s_t^2 + \beta_3' \tilde{\mathrm{x}}_t s_t^3 + v_t$
  - $\hat{\varepsilon}_t$: the residuals of the linear AR; $\tilde{\mathrm{x}}_t$: x_t without the constant; β_k: the coefficients of the interactions with s_t^k; v_t: the error
  - LM3: $\beta_1 = \beta_2 = \beta_3 = 0$, an F test with $3p$ restrictions; repeat for each candidate s_t and take the smallest p-value (derivation in the Appendix)
- **Choice of the family** Teräsvirta (1994): test H_{04}: $\beta_3 = 0$, then H_{03}: $\beta_2 = 0 \mid \beta_3 = 0$, then H_{02}: $\beta_1 = 0 \mid \beta_2 = \beta_3 = 0$
  - strongest rejection of H_{03}: ESTAR; of H_{04} or H_{02}: LSTAR (an ESTAR has no cubic term in its expansion)
- Estimation by nonlinear least squares, linear parameters concentrated out; evaluation by LM tests of no remaining nonlinearity and parameter constancy Eitrheim and Teräsvirta (1996)

Case study: van Dijk, Teräsvirta and Franses (2002)

- The paper, Section 7: unadjusted unemployment rate of US men aged 20+, June 1968–December 1999; estimation to 1989, forecasts for 1990–1999 van Dijk, Teräsvirta and Franses (2002)
  - Δy_t on a constant, 11 monthly dummies, y_{t-1} and 15 lags of Δy_t; transition variable $s_t = \Delta_{12} y_{t-d}$ (the 12-month change, lagged d months), $d = 1, \ldots, 6$
  - their LSTAR with $d = 1$: $\hat{\gamma} = 23.15$, $\hat{c} = 0.27$, residual s.d. 0.92 of the AR
- Our replication: the same series (BLS via FRED), sample, regressors and transition; the unrestricted lags, without their pruning of insignificant terms
- Question: does an LSTAR beat the linear model out of sample?

An LSTAR for US unemployment



- Left: the estimated transition function over time with the unemployment rate; right: G as a function of $s_t = \Delta_{12} y_{t-1}$, ours and the published one

ATS_ch2_unemp_lstar

Interpreting the LSTAR

- LM3 p-values: $d = 1$: 0.124, $d = 2$: 0.052, $d = 3$: 0.275; for $d = 2$: H_{04} 0.008, H_{03} 0.548, H_{02} 0.461: LSTAR
  - in line with van Dijk, Teräsvirta and Franses (2002): weak evidence, strongest for $d = 2$, pointing to the logistic family
- $\hat{\gamma} = 21.0$, $\hat{c} = 0.30$, residual s.d. ratio 0.90; the regime switches when unemployment has risen by about 0.3 pp in a year ($G > 0.5$ in 32% of months)
  - AIC prefers the LSTAR (-3.034 against -2.979); BIC the linear model (-2.573 against -2.366): 18 extra parameters without pruning
- 1990–1999, one step: RMSE 0.218 (LSTAR) against 0.228 (AR), HLN 1.14, p 0.255: better, not significantly

Real exchange rates and purchasing power parity



Gustav Cassel (1866–1945)

Portrait: unknown author; public domain; Wikimedia Commons



Bretton Woods, July 1944

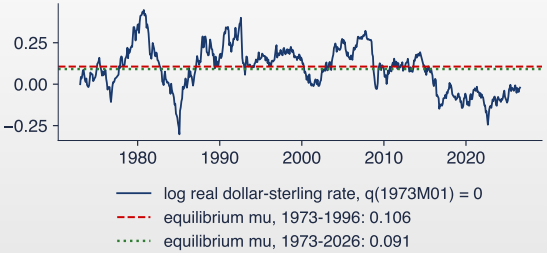
Photo: US National Archives (1944); public domain; Wikimedia Commons

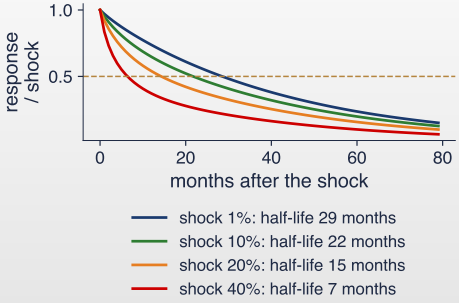
- Purchasing power parity, PPP (Cassel (1918)): the real exchange rate $q_t = s_t - p_t + p_t^*$ should be stationary
  - s_t: the log nominal rate (domestic per foreign currency); p_t, p_t^*: log domestic and foreign price levels
  - after Bretton Woods ended (1973) unit-root tests rarely reject
  - first PPP puzzle: no mean reversion; second: half-lives of 3–5 years, too slow for nominal shocks
- Transaction costs create a band of inaction: near parity q_t is close to a random walk, far from it arbitrage pulls it back Michael, Nobay and Peel (1997)
  - Taylor, Peel and Sarno (2001): ESTAR $q_t - \mu = (q_{t-1} - \mu)\exp\{-\theta^2(q_{t-1} - \mu)^2\} + \varepsilon_t$
  - μ: the equilibrium; near it the AR coefficient $\exp\{\cdot\}$ is close to 1 (random walk), far from it close to 0 (fast reversion); θ^2: the speed

Case study: Taylor, Peel and Sarno (2001)

- The paper: monthly real dollar rates of sterling, mark, franc and yen, 1973M01–1996M12, IMF IFS data, $q(1973M01) = 0$; ESTAR with $p = d = 1$ and $\beta_1 = -\beta_1^* = 1$ (eq. 8) Taylor, Peel and Sarno (2001)
  - dollar–sterling: $\hat{\theta}^2 = 0.452$, $\hat{\mu} = -0.149$, $s = 0.033$ (residual standard deviation); Monte Carlo p-value of $\hat{\theta}$ under a random walk: 0.002
  - half-lives from generalised impulse responses conditional on the average history: under one year for a 40% shock, just under three years for 1%
- Our replication for dollar–sterling: end-of-month DEXUSUK, US CPI (CPIAUCNS), UK CPI (OECD to 1987, then ONS); same model, same half-life procedure; extension to August 2026
- Question: is the real exchange rate nonlinearly mean reverting, and how fast?

Nonlinear mean reversion of the real dollar–sterling rate





⊡ Left: q_t and the estimated equilibria; right: generalised impulse responses of the 1973–1996 ESTAR, shocks of 1–40%, averaged over the observed histories

ATS_ch2_ppp_estar

Interpreting the ESTAR

- 1973–1996: $\hat{\theta}^{\mathbf{2}} = 0.527$ (s.e. 0.219), $\hat{\mu} = 0.106$, $s = 0.033$; Monte Carlo p-value of the t-ratio 0.165 (paper: 0.002)
  - $\theta^{\mathbf{2}}$ and s are close to the paper; μ differs because the UK CPI of the paper (IFS) is not the series published today
- Half-lives: 1% shock 29 months, 10% 22, 40% 7; linear AR(1): 21 months; from equilibrium: 42 months for 1%
  - large deviations die out fast: the second PPP puzzle is a feature of small shocks
- Dickey–Fuller: -2.20 (p 0.205); its power against our ESTAR is 19%
- KSS Kapetanios, Shin and Snell (2003): -2.47 (p 0.144) to 1996, -3.47 (p 0.014) to 2026
  - the t statistic of δ in $\Delta q_t = \delta q_{t-\mathbf{1}}^{\mathbf{3}} + e_t$ (demeaned q_t): a unit-root test against a stationary ESTAR; reject in the left tail

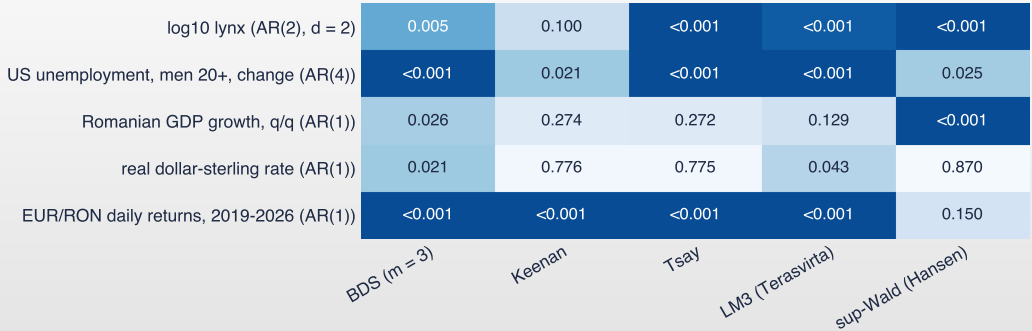
Recap: Smooth transition

- LSTAR: asymmetry between low and high states; ESTAR: small and large deviations
- LM tests from a Taylor expansion avoid the identification problem; the sequence H_{04}, H_{03}, H_{02} chooses the family
- Unemployment: the published LSTAR replicates; its forecast gain is small
- PPP: half-lives depend on the size of the shock; the evidence for nonlinearity depends on the sample

General and specific tests

- **BDS** Brock et al. (1996): correlation integral $C_m(\varepsilon)$ of m-histories; under i.i.d. $C_m = C_1^m$; applied to the residuals of a linear model
  - $C_m(\varepsilon)$: the share of pairs of m-histories $(y_t, \ldots, y_{t+m-1})$ that are within distance ε in every coordinate
  - power against any dependence left in the residuals, including ARCH: a rejection does not say which nonlinearity
- **Keenan** (squared fitted values) and the test of Tsay (1986) (all cross products $y_{t-i}y_{t-j}$, orthogonalised): F tests of second-order Volterra terms
  - Tsay's arranged-autoregression test targets the TAR alternative Tsay (1989)
- **LM3** (STAR) and **sup-Wald** (TAR): specific alternatives, more power when they are right
  - caution: structural breaks, outliers and heteroskedasticity also reject linearity Carrasco (2002)

Nonlinearity tests on five series



- p-values; darker: stronger rejection of linearity. BDS on the AR residuals ($m = 3$, $\varepsilon = \hat{\sigma}$); LM3 and sup-Wald with $s = q = y_{t-d}$; sup-Wald with 200 bootstrap replications

ATS_ch2_nonlinearity_tests

Interpreting the test battery

- Lynx and unemployment: every specific test rejects; the nonlinearity is real and of the threshold type
- EUR/RON returns: BDS, Keenan, Tsay and LM3 all reject with $p < 0.001$, yet the TAR sup-Wald does not (p 0.150)
  - volatility clustering and outliers, not a nonlinear conditional mean: the heteroskedasticity-robust test is the honest one
- Romanian GDP growth: only BDS and sup-Wald reject ($p < 0.001$): the 2009 and 2020 quarters act as a "regime"; a break or an outlier model may explain the same data

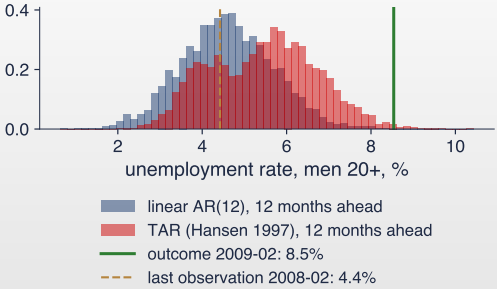
Threshold cointegration

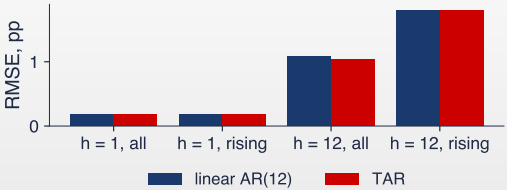
- Balke and Fomby (1997): the error-correction term $z_t = y_t - \beta x_t$ adjusts only outside a band: $\Delta z_t = \rho\, z_{t-1} 1\{|z_{t-1}| > \gamma\} + \varepsilon_t$
  - y_t, x_t: two cointegrated prices; β: the cointegrating coefficient; $\rho < 0$: the speed of adjustment outside the band $[-\gamma, \gamma]$
  - the same transaction-cost argument as ESTAR, for a pair of prices: commodity prices in two markets, interest-rate pass-through, the law of one price
- Asymmetric adjustment (momentum TAR) Enders and Siklos (2001); the test of linear against threshold VECM with a bootstrap Hansen and Seo (2002)
  - Chapter 4 introduces the linear VECM and ARDL; threshold versions are a natural project extension
- Romanian example: pass-through from the BNR policy rate to ROBOR and to lending rates, with a band where banks do not adjust

Multi-step forecasts and the model skeleton

- For $h = 1$: $\mathbb{E}(y_{t+1} \mid \mathcal{F}_t) = F(\mathrm{y}_t)$; for $h \geq 2$: $\mathbb{E}[F(F(\mathrm{y}_t) + \varepsilon_{t+1}, \dots)] \neq F(F(\mathrm{y}_t))$ (Jensen)
  - F: the conditional mean function of the model; Jensen: the mean of a nonlinear function is not the function of the mean
  - iterating the skeleton (the "naive" forecast) is biased; simulate paths with bootstrapped residuals and average them Clements, Franses and Swanson (2004)
  - the predictive density can be skewed or bimodal: report densities, scored by the CRPS or the log score (Chapter 1)
- Evidence: nonlinear models rarely beat linear ones on average; gains concentrate in particular regimes (recessions) Montgomery et al. (1998), Teräsvirta, van Dijk and Medeiros (2005)
  - evaluate conditionally: GW tests with the regime as instrument (Chapter 1); asymmetric dynamics in US unemployment Koop and Potter (1999)

TAR and AR forecasts of US unemployment





- Hansen's TAR ($d = 12$) and a linear AR(12) in differences, parameters fixed at 1959–1996; forecasts of the level by 500 simulated paths, origins 1996–2019. Left: 12-month densities from February 2008

ATS_ch2_nonlinear_forecasts

Interpreting the nonlinear forecasts

- From February 2008 (before the 2008–2009 surge): TAR mean 5.5%, 90% range [3.5; 7.5]; AR mean 4.6%, [2.9; 6.4]; outcome 8.5%
  - the TAR density is skewed upwards once unemployment starts rising; neither model foresaw the size of the recession
- $P = 270$ months: RMSE at 1 month 0.173 (TAR) against 0.177 (AR), HLN 1.12 (p 0.265); at 12 months 1.039 against 1.087 (p 0.599)
  - in the 62 rising-regime months the two are equal at 12 months (1.802 and 1.801): no regime-specific gain out of sample
- A good in-sample nonlinearity is not a forecasting gain: pre-register the comparison

Recap: Nonlinearity tests and forecasts

- General tests (BDS, Keenan, Tsay) detect, specific ones (LM3, sup-Wald) describe
- Heteroskedasticity, outliers and breaks also reject linearity
- Multi-step nonlinear forecasts need simulation; report densities
- Out of sample, nonlinear models gain little on average; test where they should gain

An open question

- Nonlinearity or breaks? A few shifts of the mean can look like nonlinear mean reversion, and a threshold model can look like a sequence of breaks Carrasco (2002), Diebold and Inoue (2001)
  - formal: is the KSS (ESTAR) rejection for the real dollar–sterling rate robust to Bai–Perron mean regimes estimated on the same data?
  - falsified if the rejection disappears once the null distribution accounts for the estimated breaks
- Why it matters: half-lives, PPP and the credibility of exchange-rate models depend on which story is true
  - literature to start from: Taylor, Peel and Sarno (2001), Kapetanios, Shin and Snell (2003), Michael, Nobay and Peel (1997), Carrasco (2002), Bai and Perron (2003)

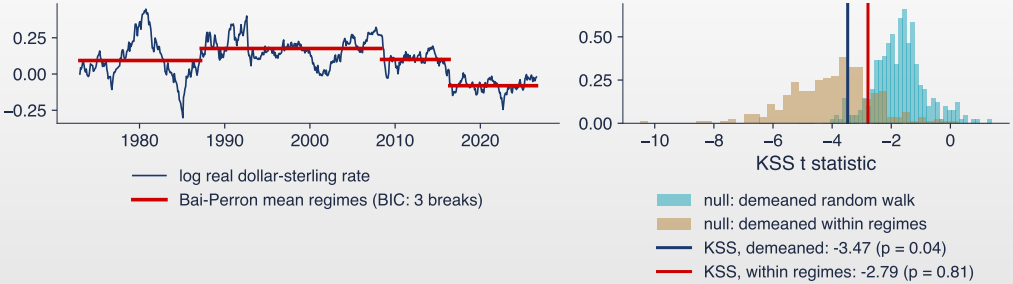
The discovery loop with an AI assistant

- An AI assistant (an LLM such as Claude, ChatGPT, Gemini or Copilot) speeds up each step; Semantic Scholar and Elicit help with the literature
  - **literature**: `List peer-reviewed papers that test ESTAR real exchange rate models allowing for breaks in the mean; give DOIs.` Then check every DOI on Crossref
  - **hypothesis**: `Propose a data-generating process with mean breaks and no nonlinearity that makes the KSS test reject.`
  - **code and replication**: reproduce first a number of this lecture (the KSS statistic to 2026), then the new procedure
  - **robustness and critique**: `Act as a referee: list every way in which estimating breaks on the same data biases the test.`
- Report: what was asked, what was kept, what was rejected (AI_USE.md, AI_ERRORS.md)

Required checks

- Every reference exists and says what is claimed (DOI resolves, title matches, the result is in the paper)
- The null distribution reproduces the whole procedure: breaks estimated on each simulated series, not fixed at the observed dates
- The number of breaks, the trimming and the criterion (BIC) are fixed before seeing the test result
- The data are the published series (vintage, deflator, end of month); differences from the paper are reported
- "Not significant within regimes" is not "linear": report the power of the procedure

Mini-case: the KSS test after Bai–Perron mean shifts



- Left: real dollar–sterling rate with Bai–Perron mean regimes (BIC: three breaks, March 1987, May 2008, May 2016); right: KSS null distributions from 300 random walks put through the same steps
- Demeaned: KSS -3.47, p 0.037 (5%: -3.13); within regimes: -2.79, p 0.813 (5%: -6.30): the evidence for ESTAR disappears once the break search is part of the null

ATS_ch2_ai_case

Project idea

- **Regimes in Romanian inflation and the policy rate**: breaks, thresholds or both?
  - replicate first: the Bai–Perron dates of this lecture and the TAR test of Hansen (1997) on the US data
  - extension: an AR model of Romanian inflation with a partial break in the intercept against an LSTAR in the 12-month change of inflation; pre-register the comparison (1–12 months, CRPS, Holm (1979))
  - optional: threshold pass-through from the BNR rate to ROBOR (Chapter 4)
- Deliverables follow the course rules: repository, report, AI_USE.md, AI_ERRORS.md, oral defence

Key takeaways

- A break date chosen from the data is a nuisance parameter: use sup/exp/ave tests with their own critical values
- Bai–Perron: global LS by dynamic programming, UDmax and sequential tests, BIC/LWZ, intervals for the dates
- Monitor with a CSW boundary; test variance breaks with κ_2; bootstrap unit-root tests with breaks
- TAR and STAR: test with the bootstrap or LM expansions, estimate by concentrated (N)LS, forecast by simulation
- Breaks and nonlinearity mimic each other: test one allowing for the other

Self-assessment

Questions

- Why is the 5% critical value of sup-Wald with one coefficient above 8, not 3.84?
- What does dynamic programming save in the Bai–Perron estimator?
- Why does a repeated 5% test eventually raise a false alarm?
- Why is the threshold of a TAR estimated at rate T?
- When does the half-life of an ESTAR depend on the shock?

Next: SVAR and local projections

- Structural VAR and local projections: identification of shocks
- State-dependent local projections extend this chapter's nonlinear models to impulse responses

Appendix: the limit of the sup-Wald statistic

- Mean shift, σ^2 known: $W_T(\pi) = \dfrac{T\pi(1-\pi)(\bar{y}_1 - \bar{y}_2)^2}{\sigma^2}$, $\bar{y}_1$, $\bar{y}_2$ the means before and after $[\pi T]$
- With $S_T(r) = \sigma^{-1} T^{-1/2} \sum_{t \le [rT]} (y_t - \mu) \Rightarrow B(r)$ (FCLT, Chapter 0):
  $\bar{y}_1 - \bar{y}_2 = \dfrac{\sigma}{\sqrt{T}} \left[\dfrac{S_T(\pi)}{\pi} - \dfrac{S_T(1) - S_T(\pi)}{1-\pi} \right]$
- The bracket equals $\dfrac{S_T(\pi) - \pi S_T(1)}{\pi(1-\pi)}$, so $W_T(\pi) \Rightarrow \dfrac{[B(\pi) - \pi B(1)]^2}{\pi(1-\pi)}$, a squared Brownian bridge over its variance
- The continuous mapping theorem gives the limit of $\sup_{\pi \in \Pi}$; with p coefficients, B is p-dimensional; with an unknown σ^2 or a HAC variance the limit is the same

Back

Appendix: the critical value 7.35 of the threshold LR

- Hansen (2000): with a shrinking threshold effect, $\mathrm{LR}_T(\gamma_0) \to_d \eta^2 \xi$, $\xi = \max_s[2W(s) - |s|]$, W a two-sided Brownian motion
- For one side, $\max_{s \geq 0}[2W(s) - s]$ is exponential with mean 2: $P(\cdot \leq x) = 1 - e^{-x/2}$ (the maximum of a Brownian motion with drift)
- The two sides are independent: $P(\xi \leq x) = (1 - e^{-x/2})^2$; solving $(1 - e^{-x/2})^2 = 1 - \alpha$ gives $c(\alpha) = -2\ln(1 - \sqrt{1 - \alpha})$
- $c(0.10) = 5.94$, $c(0.05) = 7.35$, $c(0.01) = 10.59$; under heteroskedasticity divide LR by $\hat{\eta}^2$ before comparing

Back

Appendix: the Taylor expansion behind LM3

- LSTAR: $G(s;\gamma,c) - \frac{1}{2} = \frac{1}{4}\gamma(s-c) - \frac{1}{48}\gamma^3(s-c)^3 + O(\gamma^5)$ around $\gamma = 0$ (with $\hat{\sigma}_s = 1$)
- Substituting in $\phi_1' x_t + (\phi_2 - \phi_1)' x_t G$ and collecting powers of s_t gives $\beta_0' x_t + \sum_{k=1}^{3} \beta_k' \tilde{x}_t s_t^k$, every β_k proportional to γ
- So H_0: $\gamma = 0$ becomes $\beta_1 = \beta_2 = \beta_3 = 0$, testable without estimating c; the first-order expansion (LM1) has no power when only the intercept switches Luukkonen, Saikkonen and Teräsvirta (1988)
- ESTAR: $1 - e^{-\gamma(s-c)^2} = \gamma(s-c)^2 + O(\gamma^2)$: squares only, hence the decision rule of Teräsvirta (1994)

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